Generative Adversarial Network-Based Full-Space Domain Adaptation for Land Cover Classification From Multiple-Source Remote Sensing Images

Shunping Ji, *Member, IEEE*, Dingpan Wang, and Muying Luo

***Abstract*— The accuracy of remote sensing image segmentation and classification is known to dramatically decrease when the source and target images are from different sources; while deep learning-based models have boosted performance, they are only effective when trained with a large number of labeled source images that are similar to the target images. In this article, we propose a generative adversarial network (GAN) based domain adaptation for land cover classification using new target remote sensing images that are enormously different from the labeled source images. In GANs, the source and target images are fully aligned in the image space, feature space, and output space domains in two stages via adversarial learning. The source images are translated to the style of the target images, which are then used to train a fully convolutional network (FCN) for semantic segmentation to classify the land cover types of the target images. The domain adaptation and segmentation are integrated to form an end-to-end framework. The experiments that we conducted on a multisource data set covering more than 3500 km^2 with 51 560 256 × 256 high-resolution satellite images in Wuhan city and a cross-city data set with 11 383 256 × 256 aerial images in Potsdam and Vaihingen demonstrated that our method exceeded the recent GAN-based domain adaptation methods by at least 6.1% and 4.9% in the mean intersection over union (mIoU) and overall accuracy (OA) indexes, respectively. We also proved that our GAN is a generic framework that can be implemented for other domain transfer methods to boost their performance.**

***Index Terms*— Domain adaptation, generative adversarial network (GAN), image classification, remote sensing.**

I. Introduction

DEEP learning has made spectacular breakthroughs in many computer vision tasks, such as semantic segmentation [1]–[3] and object detection [4], which are driven by the progress of deep convolutional neural networks (CNNs) and large quantities of available training data sets. Unfortunately, CNNs rarely generalize learned knowledge to new data sets, especially when there is a wide domain gap between the source and target images. For example, there are usually significant appearance differences between the remote sensing images acquired from different satellites. It is never expected that a pretrained model can acquire satisfactory performance when applied directly to another quite different remote sensing data set, at least in the current stage. Hence, labeled samples in each upcoming satellite image are required to train a new deep learning-based model. Although a large number of historical GIS maps are available, these vector maps are outdated soon. As a result, an extensive amount of manual work is applied to update vector maps, which can then be provided as up-to-date training samples. The other problem is the access to updated high-quality vector maps is not easy. To address these problems and make the best of the available vector maps, unsupervised domain adaptation (UDA) has been proposed to reduce the domain gap between the source and target data to enable a pretrained deep learning model to achieve competitive results on unlabeled target data.

Recently introduced methods have focused on cross domain semantic segmentation from high-quality close-range or medical images [5]–[10]. For example, Hoffman *et al.* [5] introduced a generative adversarial network (GAN)-based feature space domain adaptation using adversarial training to align the CNN features from the source and target domains to improve the segmentation performance of target images, where only the source images had labeled samples. Following this approach, further studies addressed the domain transfer in semantic segmentation using different GAN-based adversarial training techniques in different space dimensions, such as the combination of image translation techniques (image space domain adaptation) and feature alignment (feature space domain adaptation) for lane image segmentation [6], image synergizing and feature adaptation for medical image segmentation [7], and synthetic-to-real adaptation at the output space domain [8]–[10].

Very recently, several GAN-based UDA methods have been shown to have some effects on processing remote sensing images obtained from different data sources. However, this area of research is still in its initial stages, and more sophisticated methods are expected to be developed for handling remote sensing data that are more complex than close-range data. For example, the work in [11] was only an application of CycleGAN in remote sensing image transfer at the image space. Another method [12] named ColorMapGAN

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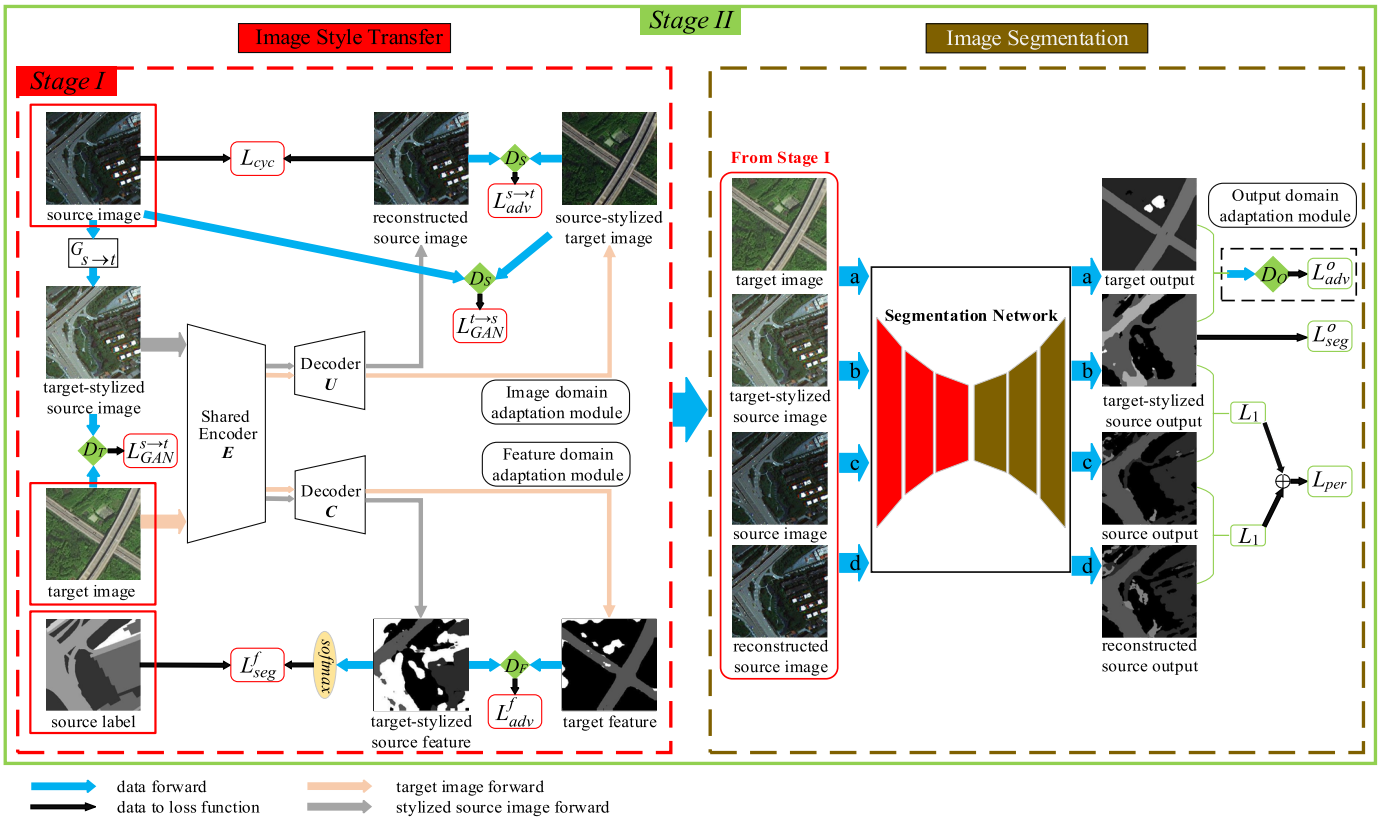


Fig. 1. Overview of our FSDAN architecture. Stage I, i.e., image style transfer, consists of an image space domain adaptation module and a feature space domain adaptation module. It takes the three images enclosed by red boxes as input and outputs target-stylized source image and reconstructed source image for image segmentation. The image segmentation consists of a segmentation network and an output space domain adaptation module. In training, it takes the four images in the red box as input; in the test, it takes only the target image as input. $\{D_S, D_T, D_F, D_O\}$ represents the discriminators, and L represents the various loss functions. {a, b, c, d}, which are in the blue arrows, indicate that each input image goes through the segmentation network, respectively.

attempted to learn the color transformation from source images to target images. Both approaches stay in the image space domain transfer without touching the more advanced feature space and output space domain adaptation.

In this article, we propose a novel end-to-end GAN-based full-space domain adaptation framework for land cover classification of the unlabeled target remote sensing images with labeled source data, i.e., existing remote sensing images with corresponding GIS data. The core idea of this approach is using the adversarial learning in all of the image space, feature space, and output space to transfer the source domain to the target domain and to realize the segmentation of the target images simultaneously. The proposed method consists of two stages.

Stage I: The image style transfer model aligns the distribution of the source images and target images both in the image space and the feature space.

Stage II: The output space domain transfer model further reduces their distribution gaps between the output spaces, and the segmentation network predicts the output results.

The data are forwarded from stage I to stage II; inspired by [13] and [14], we propose a perceptual loss to build the bridge from stage II to stage I to form a closed and uniform training loop.

The main contributions of this article are as follows. First, we propose a novel end-to-end GAN-based full-space domain adaptation learning framework, which, to the best of our knowledge, is the first full-space domain transfer method. It is not only applicable in land cover classification from multisource remote sensing images but also can be generic to many UDA tasks. Second, we propose an image style transfer method that generates stylized images similar to the target domain images for improving the performance of deep learning tasks. Third, in its implementation in two different large-scale data sets, one consists of bitemporal satellite images in Wuhan city, and the other consists of open-source aerial image sets in Potsdam and Vaihingen; our method demonstrated superior performance for segmentation adaptation from multiple-source remote sensing images and markedly improved accuracy over current state-of-the-art methods.

II. Method

Our full-space domain adaptation network (FSDAN) is shown in Fig. 1, which is designed for the application case where a new image set without labeled samples needs to be categorized pixel-by-pixel and another set that has similar scenes but a quite different appearance possesses the available labels. This case is very common in land cover classification from remote sensing images where a large number of historical GIS maps are available. The framework consists of two stages. Stage I is composed of an image space domain adaptation module and a feature space domain adaptation module, both of which, in a collaborative manner, are utilized to narrow the gap

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between the visual appearances of two image sets and to output both the target-stylized source images and the reconstructed source images. Stage II takes both of them and the original source and target images as inputs, utilizes adversarial learning to align the spatial layout and local context of the two domains in the output space, and trains a fully convolutional network (FCN) simultaneously to predict segmentation maps of the target images. In addition, the loss functions in stage II can be backpropagated to modify the weight parameters in stage I to form a closed and uniform training loop.

In the training procedure, the inputs of the two stages are enclosed with red solid boxes, as shown in Fig. 1. In the test procedure, the input is only the target images to be fed into the segmentation network.

A. Stage I: Image Style Transfer

Stage I attempts to translate a source image set X_S with known labels Y_S, to the domain of a target image set X_T, which is lacking labels and needs to be classified. By doing this, a task model trained on the translated source data can be well adapted to the target data, and an unsupervised classification problem can be adequately supervised.

1) Image Space Domain Adaptation Module: In our image space domain adaptation, we follow the classic GAN structure proposed in [15]. In (1), the output of discriminator D_T is a binary value of 0 or 1, indicating that the discriminator judges the input images that are from the source domain or target domain, respectively. $G_{S\to T}$ is a generator aligning the data distribution of images from the source domain to the target domain; the generated results are denoted as $G_{S\to T}(x_s)$, where x_s are the input from the source domain. The loss function L_{GAN} encourages $G_{S\to T}$ to fool D_T. D_T is encouraged to discriminate the real target images x_t from the target-stylized source images $G_{S\to T}(x_s)$. Through the two iterative steps, they play against each other and evolve with the adversarial learning

$$L_{\text{GAN}}^{s\to t}(G_{S\to T}, D_T) = \mathbb{E}_{x_t\sim X_T}[\log D_T(x_t)] + \mathbb{E}_{x_s\sim X_S}[\log(1 - D_T(G_{S\to T}(x_s)))]. \quad (1)$$

Cycle consistency [16] also is imposed on the image domain transfer. As shown in Fig. 1, the encoder E generates abstracted high-dimensional features from the input x_t, and the generated features are denoted as $E(x_t)$. The decoder U projects the features, i.e., $E(x_t)$, to image space. The decoder C projects $E(x_t)$ to low-dimensional features. The encoder E and decoder U together form the reverse generator $G_{T\to S}$, i.e., $G_{T\to S}(x_t) = U(E(x_t))$, which translates the target images x_t into the source-like images $G_{T\to S}(x_t)$ (i.e., $G_{T\to S}(x_t) \approx x_s$); the discriminator D_S distinguishes x_s from $G_{T\to S}(x_t)$. According to this, another GAN loss $L_{\text{GAN}}^{t\to s}(G_{T\to S}, D_S)$ is constructed in the same way as (1). Then, the pixel-level cycle-consistency loss L_{cyc} is used to encourage the two mappings for recovering the original images, which are $G_{T\to S}(G_{S\to T}(x_s)) \approx x_s$ and $G_{S\to T}(G_{T\to S}(x_t)) \approx x_t$, respectively. The cycle-consistency loss is formulated as

$$L_{cyc}(G_{S\to T}, E, U) = \mathbb{E}_{x_t\sim X_T}\left[\|G_{S\to T}(U(E(x_t))) - x_t\|_1\right] + \mathbb{E}_{x_s\sim X_S}\left[\|U(E(G_{S\to T}(x_s))) - x_s\|_1\right]. \quad (2)$$

Equations (1) and (2) are similar to those used in CycleGAN [16]. However, our framework has a shared encoder E for both decoder C and decoder U. The parameters of E are modified by the segmentation loss and adversarial loss in stage II. To balance the backpropagation gradient between E and $G_{S\to T}$, we propose an additional adversarial loss $L_{\text{adv}}^{s\to t}$, which is added to discriminate whether the generated images are transformed from real target images x_t or reconstructed from $G_{S\to T}(x_s)$. The discriminator D_S attempts to discriminate the reconstructed source images from the source-stylized target images, as shown in the following equation:

$$L_{\text{adv}}^{s\to t}(G_{S\to T}, D_S) = \mathbb{E}_{x_t\sim X_T}\left[\log D_S(U(E(x_t)))\right] + \mathbb{E}_{x_s\sim X_S}\left[\log(1 - D_S(U(E(G_{S\to T}(x_s)))))\right]. \quad (3)$$

2) Feature Space Domain Adaptation Module: The image space domain adaptation module focuses on the transformation between the source and target images at the image-level. However, it is not an easy task for a GAN-based module to learn complete transferable knowledge and achieve the desired image style transfer, especially when the domain shift is complicated. Inspired by SIFA [7], we developed a collaborative training framework for image style transfer that transforms the appearance of the images across the domains both in image space and feature space, where the feature space domain adaptation then further enhances the domain-invariance of the extracted CNN features.

As multilayer high-dimension features are difficult to align, in SIFA [7], low-dimensional features are extracted by compressing the high-dimensional features, i.e., the output of encoder E. We borrowed the idea of the shared encoder from SIFA [7]; however, in that work, the compression is implemented by a simple rough decoder C, which is composed of a single convolutional layer and a direct $16\times$ upsampling. Inspired by Deeplabv3+ [17], we added three upsampling blocks that gradually recover the spatial information to capture finer and sharper object boundaries. The structure of decoder C is described in Section II-D.

The target-stylized source images $G_{S\to T}(x_s)$ and original source images are considered as sharing the same labels y_s, which can be used as a constraint to train features. The features from $\{E, C\}$ can be converted to labels (segmentation maps) via the softmax function σ, i.e., $\sigma(C(E(G_{S\to T}(x_s))))$. We applied the segmentation loss as in SIFA [7] to optimize the module of the feature extraction, which is defined as

$$L_{\text{seg}}^{f}(E, C) = -\mathbb{E}_{(x_s,y_s)\sim(X_S,Y_S)}\sum_{k=1}^{K} 1_{[k=y_s]} \times \log(\sigma(C(E(G_{S\to T}(x_s))))) \quad (4)$$

where K is the number of categories and $1_{[]}$ is the indicator function.

Equation (4) only learns how to adjust features without domain adaptation. We applied adversarial learning [5] to achieve the feature space domain adaptation. The discriminator D_F attempts to discriminate whether the features are from target images x_t or target-stylized source images $G_{S\to T}(x_s)$. Successful discrimination of x_t and $G_{S\to T}(x_s)$ by D_F means

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that pair $\{E, C\}$ has not reached the feature-level adaptation. Then, we can use an adversarial loss to update $\{E, C\}$. The adversarial loss is

$$L^{f}_{\text{adv}}(E, C, D_F) = \mathbb{E}_{x_t \sim X_T}[\log(1 - D_F(C(E(x_t))))] + \mathbb{E}_{x_s \sim X_S}[\log D_F(C(E(G_{S\to T}(x_s))))]. \quad (5)$$

B. Stage II: Output Space Domain Transfer

Although (4) shows a segmentation map of the target images created, the map is rough and cannot be treated as the final results. On the other hand, although we obtained target-stylized source images from stage I, a domain shift may still exist, as [8] and [18] reported, because the adaptation in the feature space is not the optimal choice for semantic segmentation. The output space domain alignment feature can be useful supervision to guide finer training. In this section, we align the target-stylized source images and target images in the output space; in other words, we align the output maps that are produced by the same segmentation network and adjust the output maps of target images to be the final segmentation maps.

1) Output Space Domain Adaptation Module: Output space domain adaptation, which was first mentioned in [8], is based on the observation that the segmentation results of the source and target images usually share a significant amount of similarities in the spatial layout and local context. However, Tsai *et al.* [8] only tested the output space alignment between the SYNTHIA data set [19] and the Cityscapes data set [20], both of which have high-quality close-range images and relatively simple segmentation maps. Alignment only at the output space may not be suitable for remote sensing images due to the relatively rougher image quality, complex land covers, and nonlinear radiometric changes between two data sets. In our method, we integrate output space domain adaptation into the image and feature space domain adaptation in stage I.

The basic idea for output space domain adaptation consists of a segmentation network, which is supervised with labels in the source domain and predicts the segmentation outputs of both the source and target images, and an adversarial learning module, which aligns the two outputs. In Fig. 1, the target-stylized source images $G_{S\to T}(x_s)$ and reconstructed source images $U(E(G_{S\to T}(x_s)))$ generated from stage I, as well as the source images x_s and target images x_t, are separately sent to a multiscale segmentation network, which is represented as S. The first task is to train the segmentation network with target-stylized source images $G_{S\to T}(x_s)$ and labels y_s. The segmentation loss is formulated as (6) where the features have been converted to outputs with a *softmax* function

$$L^{o}_{\text{seg}}(S) = -\mathbb{E}_{(x_s, y_s)\sim(X_S, Y_S)} \sum_{k=1}^{K} 1_{[k=y_s]} \log(S(G_{S\to T}(x_s)))). \quad (6)$$

In addition, we applied adversarial learning to align the distribution of the target-stylized source images and the target images in the output space by making full use of the labels of the target-stylized source images. Our aim is that the segmentation network S also learns the knowledge to fool the discriminator D_O that attempts to discriminate the target images and target-stylized source images by using an adversarial loss

$$L^{o}_{\text{adv}}(S, D_O) = \mathbb{E}_{x_t \sim X_T}[\log(1 - D_O(S(x_t)))] + \mathbb{E}_{x_s \sim X_S}[\log D_O(S(G_{S\to T}(x_s)))]. \quad (7)$$

Up to this point, the data flow is forward-propagated from stage I to stage II. We needed gradient propagation from updated stage II back to stage I to refine the latter. Therefore, we introduced a new loss function called the perceptual loss L_{per}. The original perceptual loss in [13] and [14] was used to measure the distance of the features. Inspired by this idea, we proposed our perceptual loss to maintain the semantic consistency between the original images and stylized images. Although a stylized image and an original image may exhibit some difference in visual appearance, regardless of whether the images were stylized or not, they should have the same semantic information outputted from the segmentation networks, which would lead to our perceptual loss to tie stage I and stage II together. Especially, our perceptual loss consists of two parts: 1) the gradient from updated stage II modifies the parameters of generator $G_{S\to T}$ to generate more target-like source images and 2) the gradient also forces generator $\{E, U\}$ to output the reconstructed source images approaching to the source images x_s. Both of them, under the rule of L_1 minimum, comprise the perceptual loss

$$L_{\text{per}}(G_{s\to t}) = \lambda_{\text{per}}\mathbb{E}_{x_s\sim X_S}[\|S(G_{S\to T}(x_s)) - S(x_s)\|_1] + \lambda_{\text{per_recon}}\mathbb{E}_{x_s\sim X_S}[\|S(U(E(G_{S\to T}(x_s)))) - S(x_s)\|_1] \quad (8)$$

where $\{\lambda_{\text{per}}, \lambda_{\text{per_recon}}\}$ are the tradeoff parameters.

C. Full-Space Domain Adaptation

Finally, we present our full-space domain adaptation by combining all the modules together. The entire training process is divided into stage I and stage II. In stage I, according to the direction of the data flow, the modules are sequentially updated in the following order: $G_{S\to T} \to D_T \to E \to C \to U \to D_S \to D_F$. Especially, the generator $G_{S\to T}$ is updated first to produce the target-stylized source images. Then, discriminator D_T is updated to discriminate against the target-stylized source images $G_{S\to T}(x_s)$ from the target images x_t. Next, E and C are updated to extract the features, followed by updating U to generate the source-stylized target images and reconstruct the source images. Finally, the discriminator D_S and D_F are updated to discriminate against the source-stylized target images and the source images and the target features and target-stylized source features, respectively. The overall loss function L_{I} is as follows:

$$\begin{aligned} L_I = {} & \lambda^{s\to t}_{\text{GAN}} L^{s\to t}_{\text{GAN}}(G_{S\to T}, D_T) + \lambda^{t\to s}_{\text{GAN}} L^{t\to s}_{\text{GAN}}(E, U, D_S) \\ & + \lambda_{cyc} L_{cyc}(G_{S\to T}, E, U) + \lambda^{f}_{\text{seg}} L^{f}_{\text{seg}}(E, C) \\ & + \lambda^{s\to t}_{\text{adv}} L^{s\to t}_{\text{adv}}(G_{S\to T}, D_S) + \lambda^{f}_{\text{adv}} L^{f}_{\text{adv}}(E, C, D_F). \end{aligned} \quad (9)$$

In stage II, we use the target-stylized source images from stage I to update the segmentation network S and then to update discriminator D_O to classify the outputs generated from the target-stylized source images or target images. Finally, the generators in stage I are updated by the perceptual loss

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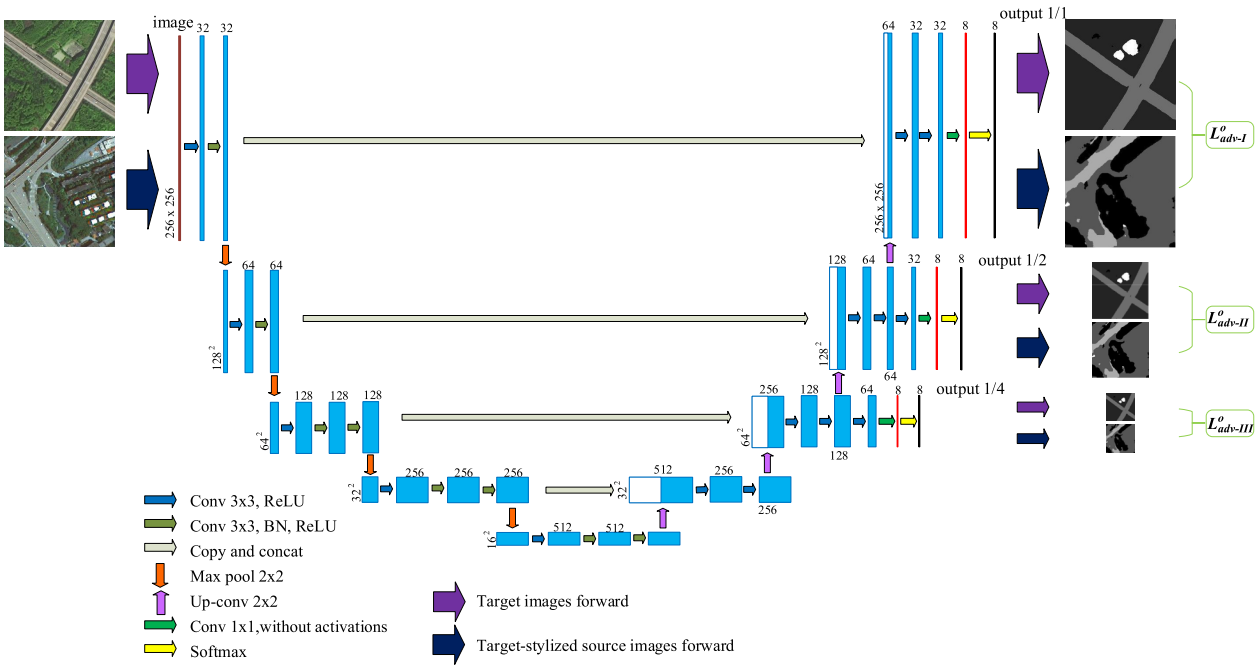


Fig. 2. Network structure that we adopted in semantic segmentation and output space domain adaptation, which is revised from MA-FCN [25]. $L^{O}_{\text{adv}-\text{I}}$, $L^{O}_{\text{adv}-\text{II}}$, and $L^{O}_{\text{adv}-\text{III}}$ are measured in the ratio of {1/1, 1/2, 1/4} of the original image size, respectively, and their addition represents the adversarial loss L^{O}_{adv} in Fig. 1.

L_{per} from stage II; stage I and II form a closed loop; and an epoch is finished. The overall loss function L_{II} is as follows:

$$L_{\text{II}} = \lambda^{o}_{\text{seg}} L^{o}_{\text{seg}}(S) + \lambda^{o}_{\text{adv}} L^{o}_{\text{adv}}(S, D_O) + L_{\text{per}}(G_{S\to T}) + L_{\text{per}}(E, U) \quad (10)$$

where $\{\lambda^{s\to t}_{\text{GAN}}, \lambda^{t\to s}_{\text{GAN}}, \lambda_{cyc}, \lambda^{f}_{\text{seg}}, \lambda^{s\to t}_{\text{adv}}, \lambda^{f}_{\text{adv}}, \lambda^{o}_{\text{seg}}, \lambda^{o}_{\text{adv}}\}$ are the tradeoff parameters to adjust the importance of each component. Since stage II has updated $G_{S\to T}$, E, and C in stage I, it is not necessary to add L_{I} in the training of stage II.

We call our algorithm "full-space domain adaptation" as we synthesize the image space domain adaptation, the feature space domain adaptation, and the output space domain adaptation in stage I and stage II and then align those domains with different adversarial learning. In contrast, all other current GAN-based domain adaptation methods align the distribution in one or two of the three spaces.

D. Networks and Configuration

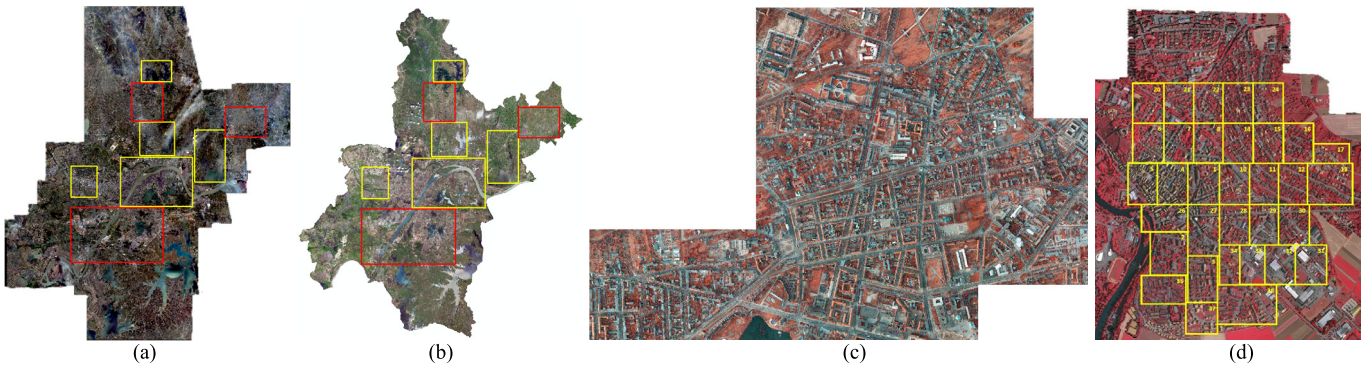
1) Network Architecture: In stage I, the structure of generator $G_{S\to T}$ follows CycleGAN [16], which consists of three convolutional layers, nine residual blocks, two deconvolutional layers, and one convolutional output layer to obtain the target-stylized source images. The structures of E and U follow SIFA [7]. Encoder E is comprised of the residual connections and dilated convolutions (dilation rate $= 2$) to enlarge the model's field-of-view while preserving the spatial information for dense predictions [21]. Decoder U consists of one convolutional layer, four residual blocks, and three deconvolutional layers, followed by one convolutional output layer to obtain the source-stylized target images. For all four discriminators $\{D_T, D_S, D_F, D_O\}$, we follow the configuration of PatchGAN [22]. The discriminator network consists of five convolutional layers with a 4×4 kernel size and a stride of 2, except for the last two layers, which uses a stride of 1. The numbers of the feature maps are {64, 128, 256, 512, 1} for each layer. Except for the last layer, each convolution layer is followed by a leaky ReLU [23] parameterized by 0.2.

The decoder C that we designed is composed of three upsampling blocks, each of which consists of one upsampling layer by a factor of 2; two convolutional layers each with a 3×3 kernel size, a batch normalization (BN) layer [24], and an ReLU activation; and one convolutional output layer with a 1×1 kernel size and without activation. The size of the output feature is the same as the original image size. The pair $\{E, C\}$ constitute the feature extractor for feature space domain transfer.

In stage II, our segmentation network is modified from a multiscale segmentation network MA-FCN [25]. The backbone and the encoder are the same, while the difference in the decoder is that we do not upsample the output feature maps of each scale and concatenate them. Instead, the feature maps of each scale are processed by a 1×1 convolutional layer and *softmax* function, which is actually the multiscale output maps that are supervised by labels with a corresponding downsampling rate {1, 2, 4}. For output space domain adaptation, we use multilevel adversarial learning [8], i.e., the final L^{o}_{adv} is the addition of scales I, II, and III as shown in Fig. 2.

2) Hyperparameter Configuration: Decoder C and MA-FCN are trained using Adam, an adaptive moment estimation optimizer with a learning rate of 2e-4, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon =$ 1e-8, and a stepped learning rate decay rate of 0.9 every two epochs. For other networks, we use Adam with $\beta_1 = 0.9$ with the other superparameters remaining unchanged. According to our empirical trial, we set $\lambda_{\text{per}} = 0.1$ and $\lambda_{\text{per_recon}} = 10$ in (8), $\lambda^{s\to t}_{\text{GAN}} = \lambda^{t\to s}_{\text{GAN}} = \lambda_{cyc} = \lambda^{f}_{\text{seg}} = 1$

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(a) (b) (c) (d)

Fig. 3. Left two images (a) and (b) are from the satellite data set in Wuhan. (a) was obtained in 2017 with 1-m ground resolution and (b) was obtained in 2019 with 0.5-m ground resolution. The areas in the yellow box, consisting of 25 604 256 × 256 tiles, were used for training, and the red boxes were used for validation (4000 tiles) and testing (21 064 tiles). The right two images (c) and (d) represent the airborne image sets of Potsdam with 5-cm ground resolution and Vaihingen with the 9-cm ground resolution, respectively.

and $\lambda_{\text{adv}}^{s\to t} = \lambda_{\text{adv}}^{f} = 0.1$ in (9), and $\lambda_{\text{seg}}^{o} = 1$ and $\lambda_{\text{adv}}^{o} = 0.001$ in (10). The whole algorithm is implemented with TensorFlow 1.14 on the Ubuntu system. A single Nvidia TITAN RTX with 24-GB GPU is used to train the model.

III. Experiment and Results

A. Data Set and Settings

We evaluated our algorithm on two different data sets. One consists of large-scale high-resolution bitemporal satellite images covering Wuhan. The Wuhan images acquired in 2017 were from the Beijing-1 satellite, and the images in 2019 were acquired from the Gaojing-2 satellite, as shown in Fig. 3. The whole city covers an area of more than 8000 km^2; however, some regions are shaded with clouds in the 2017 images, such as the upright and right parts. We selected several representative regions up to 3500 km^2 for the experiment. The regions in the yellow box were used for training, and the regions in red boxes were used for testing. We employed the 2017 data sets as the source domain and the 2019 data sets as the target domain. Therefore, the ground truth of the 2019 data sets was applied for only evaluating the performance of the deep learning model. The original ground resolutions of the satellite data sets in 2017 and 2019 were 1 and 0.5 m, respectively. We downsampled the 2019 data sets to 1-m ground resolution.

The other data set consists of two known open-source airborne image sets of Potsdam and Vaihingen cities, as shown in Fig. 3. The images are provided from the "2-D semantic labeling contest" of ISPRS WG III/4 [26] and have been produced into true orthophotos (TOPs). The three bands of the TOPs correspond to the near-infrared, red, and green bands delivered by the camera (the blue band is not available). The original ground resolutions of Potsdam and Vaihingen were 5 and 9 cm. We downsampled the Potsdam set to 10-cm ground resolution and employed it as the source domain and the Vaihingen set as the target domain. The two data sets, therefore, cover different data sources, satellite images and aerial images, different application scenes, overlapped (Wuhan data) and nonoverlapped (Potsdam and Vaihingen data), to comprehensively evaluate the performances of domain adaptation methods.

Due to a GPU capacity limitation, the whole image was cropped into 256 × 256 tiles. The land use and land cover types of the Wuhan data sets were divided into the following eight categories according to the national geoinformation survey of China in 2017: cultivated lands, forests and grasslands, buildings, roads, structures, construction sites, deserts and bare lands (very rare in Wuhan), and water (see Fig. 4). For the Potsdam and Vaihingen data sets, there were six categories in labels: impervious surfaces, building, low vegetation, tree, car, and clutter/background.

TABLE I

Different Space Domains Being Used for Domain Adaptation in the Compared Methods

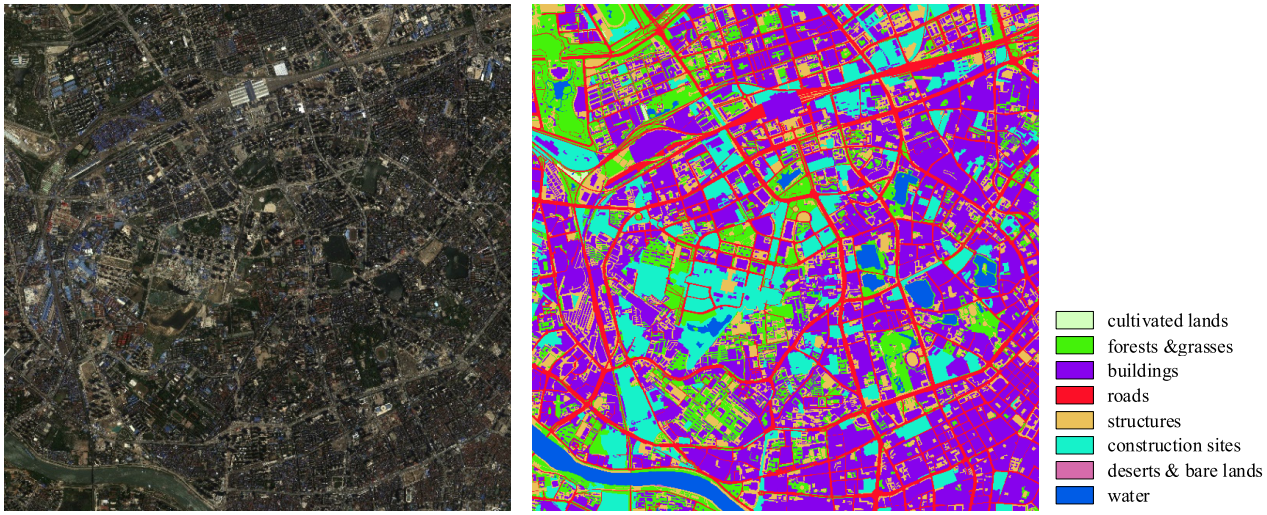
Method	Image Space	Feature Space	Output Space
CycleGAN [16]	√	×	×
ColorMapGAN [12]	√	×	×
CyCADA [6]	√	√	×
SIFA [7]	√	√	×
AdaptSegNet [8]	×	×	√
Ours	√	√	√

In the training phase, we set four 256 × 256 subimages as a minimum batch. All the data were normalized as zero mean and unit variance without any other augmentation. For Wuhan images, the training set consisted of 25 956 images in 2017 with available labels and 25 604 images in 2019 without labels. The validation and testing sets consisted of 4000 and 21 064 images in 2019, respectively. For Potsdam and Vaihingen images, the training set consisted of 4939 images in Potsdam with available labels and 6444 images in Vaihingen without labels, and the testing set consisted of 2500 images in Vaihingen.

B. Land Cover Classification

To demonstrate the advantage of our algorithm, we compared its performance to that of five current popular domain adaptation methods: CycleGAN [16], ColorMapGAN [12], CyCADA [6], SIFA [7], AdaptSegNet [8], and a baseline segmentation method MA-FCN [25]. The performance of each method is evaluated with two representative indexes: mean intersection over union (mIoU) and overall accuracy (OA). Table I summarizes which space domain or combination thereof was used for each method. Basically, except for the ColorMapGAN, the image space domain adaptation in all the other methods in Table I was based on CycleGAN.

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(a) (b)

Fig. 4. (a) Subregion of the 2017 satellite images covering 61 km^2. (b) Land cover label of (a). In (b), as shown in the legend, yellow represents cultivated lands, green represents forests and grasslands, purple represents buildings, red represents roads, orange represents structures, cyan represents construction sites, pink represents deserts & bare lands, and blue represents water.

TABLE II

SEGMENTATION RESULTS OF THE 2019 TEST DATA SET USING DIFFERENT DOMAIN ADAPTATION METHODS THAT ADAPT 2017 DATA SETS WITH LABELS TO THE 2019 DATA SETS WITHOUT LABELS. THE ACRONYMS ARE CULTIVATED LANDS (CL), FORESTS & GRASSES (FG), BUILDINGS (BD), ROADS (RD), STRUCTURES (ST), CONSTRUCTION SITES (CS), DESERTS & BARE LANDS (DB), AND WATER (WT)

2017→2019, Wuhan Data										
Method	IoU								mIoU	OA
	cl	fg	bd	rd	st	cs	db	wt		
MA-FCN (baseline)	50.2	28.2	46.3	29.4	15.0	20.8	0.1	47.8	29.7	55.8
CycleGAN	38.4	30.1	21.6	17.1	12.8	21.7	0	47.6	23.7	49.0
ColorMapGAN	49.9	36.2	30.6	27.8	14.6	20.3	0.9	44.0	28.0	55.0
CyCADA	51.7	34.1	47.1	33.9	16.9	20.5	0	55.3	32.4	58.8
SIFA	49.9	41.4	23.4	24.9	11.5	25.4	0	55.6	29.1	59.0
AdaptSegNet	51.9	36.7	48.8	34.0	14.0	31.0	0.1	61.7	34.8	61.8
FSDAN (ours)	**60.1**	**43.4**	**51.8**	**42.7**	**22.8**	**33.1**	**1.5**	**71.7**	**40.9**	**67.5**
Oracle (Trained on target)	62.1	46.9	59.3	46.7	26.4	39.8	0.2	75.6	44.6	71.0

All methods used MA-FCN (our version) as the segmentation network except SIFA, in which the feature encoder is closely coupled with the segmentation part (decoder). CycleGAN and ColorMapGAN are image style transfer networks without segmentation, so we used MA-FCN as the segmentation network to train the target-stylized source images.

Table II shows the performance of each method on the Wuhan data sets using mIoU and OA scores. According to Table II, our method considerably outperformed the second-best method AdaptSegNet by 6.1% for mIoU and 5.7% for OA; our method also achieved the highest mIoU scores in every single class. Compared with the baseline MA-FCN without domain adaptation, our method achieved 11.2% mIoU and 11.7% OA improvement, which demonstrates the importance of UDA in the unsupervised classification case. The results of our method even approach the "oracle case," which means a fully supervised case that the training set of 2019 is available.

Table III provides the segmentation results of different methods on the Potsdam&Vaihingen data set. Again, our method achieved the highest mIoU and OA scores. It outperformed the second-best method by 8.0% for mIoU and 4.9% for OA, and our method also achieved the highest mIoU scores in every single class. Besides, compared with all the other adaptation methods, our method can significantly improve the IoU score of some rare categories, e.g., car. These showed that our method is high-performing and stable to boost the segmentation results from different data sets. However, it is observed that there is still a large gap between the performance of our UDA method and the "oracle case." It can be explained that the different scenes, foreground and background, from a new city challenge a pretrained model.

We were slightly surprised that SIFA and CyCADA performed even worse than the MA-FCN in mIoU even though they have been demonstrated effective for natural or medical image data sets. The reason for these results may exist in the difference between close-range natural images and remote sensing images, the latter having much more complex scenes and usually worse image quality due to the heavy and ever-changing atmosphere. CycleGAN performed the worst on the

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TABLE III

SEGMENTATION RESULTS OF THE VAIHINGEN TEST DATA SET USING DIFFERENT DOMAIN ADAPTATION METHODS THAT ADAPT POTSDAM DATA SETS WITH LABELS TO THE VAIHINGEN DATA SETS WITHOUT LABELS. THE ACRONYMS ARE IMPERVIOUS SURFACES (IS), BUILDING (BD), LOW VEGETATION (LV), TREE (TE), CAR (CR), AND CLUTTER/BACKGROUND (CB)

Potsdam→Vaihingen								
Method	IoU						mIoU	OA
	is	bd	lv	te	cr	cb		
MA-FCN (baseline)	34.2	44.1	16.7	53.3	7.5	4.3	26.7	52.6
CycleGAN	41.3	45.0	11.9	56.5	9.1	3.7	27.9	54.4
ColorMapGAN	26.8	43.4	37.1	56.5	8.7	9.9	30.4	57.0
CyCADA	38.2	41.4	21.0	48.7	1.9	0	25.2	54.8
SIFA	44.9	3.8	11.4	47.4	4.6	0.4	18.8	47.9
AdaptSegNet	55.9	55.5	25.6	50.8	18.9	7.5	35.7	63.3
FSDAN (ours)	**57.4**	**57.8**	**41.7**	**58.4**	**37.0**	**10.0**	**43.7**	**68.2**
Oracle (Trained on target)	83.7	89.4	70.9	77.4	65.4	75.4	77.0	89.1

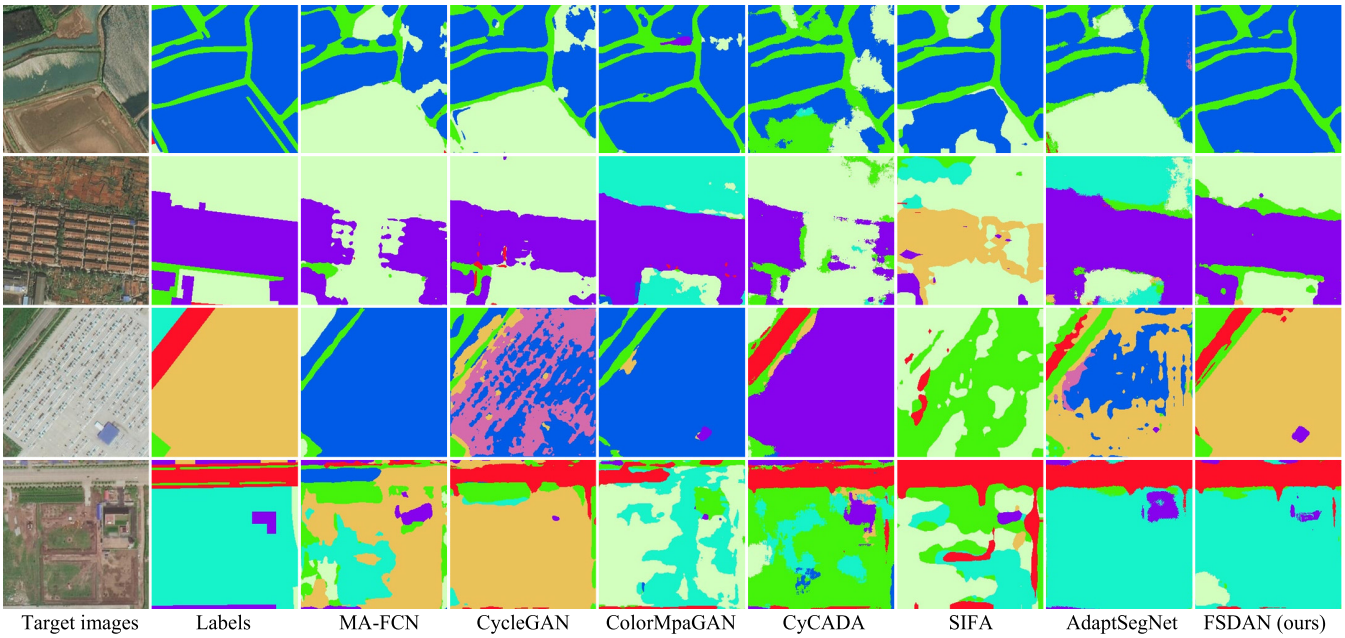


Fig. 5. Segmentation results of 256 × 256 subimages in the Wuhan data set from different methods. The colors of these image tiles represent the same categories as in Fig. 4.

Wuhan data set and did not significantly boost the performance on the Potsdam&Vaihingen data set. The performance of ColorMapGAN resembled that of the CycleGAN but was slightly better. Our observation was consistent with [27], where the image-level domain transfer resulted in worse performance in remote sensing image classification and also was somewhat consistent with [8] and [18], which reported that the feature space domain adaptation was not optimal for semantic segmentation.

In general, the output space domain transfer, i.e., AdaptSegNet, was obviously better and more robust than the feature space and image space transfer in this experiment; the other four methods with the feature- or image-level transfer were worse and unstable for different data sets; and our full space domain transfer performed best overall.

Fig. 5 shows four examples of the segmentation results for the different methods on the 256 × 256 tiles of Wuhan images. Without any adaptation, MA-FCN produced many false predictions. However, CycleGAN, ColorMapGAN, CyCADA, and SIFA did not significantly boost the performance and even made it worse in some individual tiles. In contrast, our method identified almost all types of land cover, achieving the segmentation results most similar to the labels. The second-best method was AdaptSegNet, which produced better results in third and fourth tiles than the other five methods.

Fig. 6 shows four segmentation examples of 256 × 256 subimages from the Potsdam&Vaihingen data set. MA-FCN can only identify some trees and buildings. SIFA and CyCADA made it worse in some tiles compared with MA-FCN. CorlorMapGAN, CycleGAN, and AdaptSegNet performed better than MA-FCN for some categories, i.e., impervious surfaces, building, or low vegetation. Our method achieved the best segmentation results in almost all categories.

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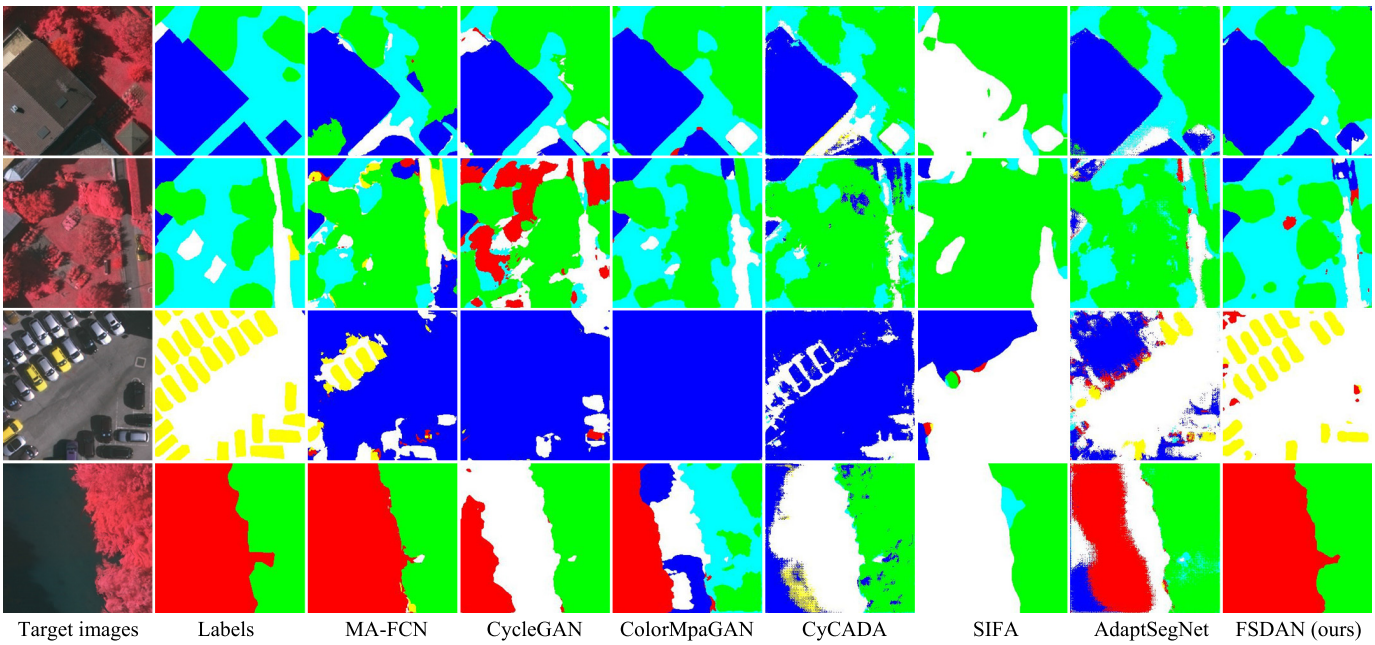


Fig. 6. Segmentation results of 256×256 subimages on the Postdam&Vaihingen data set from different methods. White: impervious surfaces; blue: building; cyan: low vegetation; green: tree; yellow: car; and red: clutter/background.

TABLE IV

ABLATION STUDY OF OUR MODEL ON THE WUHAN DATA SET. MA-FCN-I INDICATES THAT WE TREATED THE TARGET-STYLIZED SOURCE IMAGES GENERATED FROM STAGE I AS THE TRAINING SET TO TRAIN MA-FCN. MA-FCN-II INDICATES THAT MA-FCN WAS BOOSTED WITH OUTPUT SPACE DOMAIN ADAPTATION (WITHOUT STAGE I). MAFCN-I-II INDICATES THAT WE USED THE TARGET-STYLIZED SOURCE IMAGES GENERATED FROM STAGE I TO TRAIN MA-FCN-II, I.E., THE PARAMETERS OF THE TWO STAGES ARE INDEPENDENT. FSDAN IS OUR FINAL MODEL FOR WHICH THE PARAMETERS IN BOTH STAGES WERE TRAINED SIMULTANEOUSLY

Methods	Description	mIoU	OA
MA-FCN	Only segmentation	29.7	55.8
CycleGAN+MA-FCN	Image space alignment	23.7	49.0
MA-FCN-I	Stage I boost (Image and feature space alignment)	36.6	63.7
MA-FCN-II	Stage II boost (output space alignment)	36.8	63.0
MAFCN-I-II	independent Stage I and stage II	39.9	66.7
FSDAN	integrated Stage I and stage II	**40.9**	**67.5**

C. Ablation Study and Analysis

Evaluating the effectiveness and contributions of the components of a whole model is often called an ablation study in psychology and some machine learning studies. As our model is relatively complex, we conducted several experiments on the Wuhan data set to investigate how each component works.

1) Impact of Stage I: First, we analyzed the effect of applying stage I by comparing MA-FCN and MA-FCN-I of Table IV, the latter of which indicates that the source images were transferred to target-like images by stage I and MA-FCN and then applied on the transferred images. It can be seen that the introduction of stage I led to 6.9% mIoU and 7.9% OA improvement as stage I shrank the gap of the visual appearances between the source images and target images. Compared with CyCADA and SIFA in Table II, our combined image and feature space domain adaptation performed much better because SIFA uses a very rough decoder that is less successful for complicated remote sensing images, and CyCADA is not an end-to-end framework and attempts to align feature space in every high dimension, which is not a good idea. Fig. 7 provides four examples where the results of MA-FCN-I were obviously better than MA-FCN in all four tiles.

We also show the effectiveness of feature space alignment by comparing the combination of CycleGAN and MA-FCN (CycleGAN + MA-FCN) and MA-FCN-I in Table IV; the latter has exactly one more feature alignment module than the former. Results demonstrated the importance of the feature alignment, which led to 12.9% mIoU and 14.7% OA improvement.

2) Impact of Stage II: We compared MA-FCN with MA-FCN-II to show the importance of the output space domain adaptation. After integrating the output space domain adaptation to the segmentation network, the mIoU increased by 7.1%, as shown in Table IV. Accordingly, the segmentation results in the fifth column indicate that significant improvement was achieved over the third column in Fig. 7.

The effectiveness of stage I and stage II was demonstrated, respectively, in the abovementioned two experiments; their contributions were quantified, i.e., they both contributed about 7% mIoU improvement. The third experiment investigated the function of their combination (MAFCN-I-II). First, the visual

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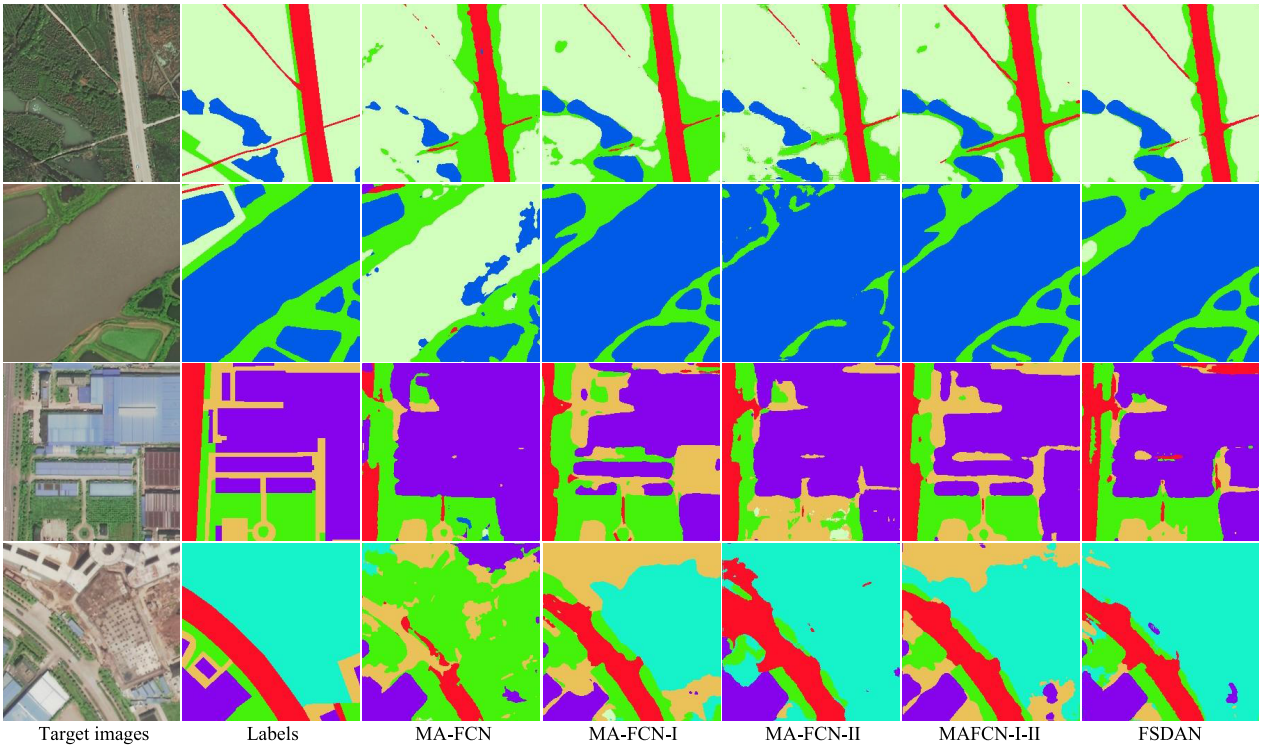


Fig. 7. Examples of segmentation results with different methods in Table IV.

appearance gap between the source domain and the target domain was reduced by the image style transfer in stage I, the output of which was target-stylized source images; then, the output domain transfer in stage II further reduced the domain gap and produced the final output. Note that there is no backward information flow from stage II to stage I. Table IV shows that the segmentation performance was improved. Compared with stage I or stage II boosted methods, the improvement with MAFCN-I-II was significant with both the mIoU and OA increasing by more than 3%.

3) Impact of Backpropagation: The fourth experiment investigated the difference between a two-step combination of image/feature level adaptation and output level adaptation (MAFCN-I-II) and a tighter combination (our final integrated FSDAN) in which the loss functions of stage II were backpropagated to modify the parameters of stage I and, subsequently, the target-stylized source images. Table IV shows that FSDAN outperformed MAFCN-I-II by 1% on mIoU and 0.8% on OA. Fig. 7 demonstrates that FSDAN was able to segment small objects more finely and drew boundaries between different land covers more precisely than MAFCN-I-II because FSDAN's backpropagation guides stage I to further decrease the visual appearance gap between the two domains and, thereby, generated target-stylized source images more similar to the target images for training a better segmentation model.

D. Generality

The method that we proposed in this article is a general full-space domain adaptation framework that can boost most of the current GAN-based domain transfer methods, which do not make full use of image space, feature space, and output space. In the abovementioned experiments, we demonstrated that our feature space and output space domain transfer can boost the image-level domain transfer because our method utilizes CycleGAN (just as all the GAN-based methods except ColorMapGAN listed in Table II) for image space domain transfer and exceeded the original CycleGAN results. In this section, we further demonstrate that using our full-space framework also can improve the feature or output domain adaptation methods. We used two very recent models as examples: 1) CLAN [10] that is a semantic consistent domain adaptation method in output space and 2) MCD [28] that is another adversarial learning method that utilizes the task-specific decision boundaries in feature space. These two models, without changing any parameter, hyperparameter, or network structure, were merged into our framework, thereby replacing the corresponding domain adaptation model.

Although CLAN produced good results on the SYNTHIA [19] and Cityscapes [20] data sets, it performed much worse on the Wuhan data set (see Table V). After it was merged into our full-space domain adaptation framework, its performance was dramatically improved by more than 10% mIoU and 20% OA. When MCD was merged into our framework, its performance was improved 4.2% mIoU score and 3% OA score from the original version.

E. Image Style Transfer

The side product of GAN-based domain adaptation for remote sensing image classification is the transferred source (or target) images, which can be the main product in computer vision, such as faked faces or mixed-style paintings. This side product is retained in our method and for other possible applications, e.g., classification with an alternative segmentation model, change detection, and object detection. Our full-space domain transfer model can produce the best quality transferred images for such applications, especially on

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TABLE V

COMPARING THE SEGMENTATION RESULTS OF USING ORIGINAL DOMAIN SPACE OF OTHER UDA METHODS AND OUR FULL SPACE ON THE WUHAN DATA SET

2017→2019, Wuhan data											
Base method	Domain space	IoU								mIoU	OA
		cl	fg	bd	rd	st	cs	db	wt		
CLAN	Output space (original)	13.6	16.3	19.9	**28.2**	8.7	14.7	**0.2**	27.7	16.1	30.6
	Full-space (ours)	**44.9**	**34.9**	**31.5**	20.2	**14.8**	**21.7**	0.0	**53.5**	**27.7**	**55.2**
MCD	Feature space (original)	**45.3**	29.4	38.9	21.4	**12.9**	18.1	**0.6**	52.9	27.4	54.2
	Full-space (ours)	39.4	**38.4**	**46.3**	**25.4**	12.8	**28.8**	0.2	**61.4**	**31.6**	**57.2**

TABLE VI

SEGMENTATION RESULTS OF WUHAN 2019 DATA USING DEEPLAB-V3+, WHICH ARE TRAINED WITH STYLIZED SOURCE IMAGES (2017) GENERATED FROM DIFFERENT STYLE TRANSFER METHODS

2017→2019, Wuhan Data											
Seg methods	Style transfer method	IoU								mIoU	OA
		cl	fg	bd	rd	st	cs	db	wt		
Deeplabv3+	CycleGAN	45.6	39.5	43.3	22.5	21.0	25.7	0.0	66.9	33.1	60.1
	SIFA	50.2	39.1	50.2	**40.7**	**23.6**	29.2	0.3	69.3	37.8	63.6
	FSDAN	**57.7**	**43.0**	**50.9**	40.1	23.1	**31.8**	**1.4**	**72.3**	**40.0**	**66.5**



Fig. 8. Examples of target-stylized source domain images generated from different methods on transferring Wuhan 2017 data to 2019 data.

pixel-level classification, which was used as an example in this study. We demonstrated this using another popular CNN-based segmentation model named Deeplabv3+ [17]. We trained Deeplabv3+ with the target-stylized source domain images generated from CycleGAN, SIFA, and our FSDAN to predict the segmentation maps of the Wuhan 2019 data set (our target data set). Table VI shows that Deeplabv3+ trained with images generated from FSDAN obtained the highest mIoU and OA scores compared with the other methods, indicating that our method generated better high-quality stylized images than the other methods.

Fig. 8 shows some examples of target-stylized source domain images corresponding to Table VI. It is observed that CycleGAN generated the worst stylized images

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with many noises. SIFA generated better stylized images; however, there are obvious noises at the lower left corner of the images. Our method generated the best-stylized images with the least noise and the clearest outlines.

IV. Discussion

The full domain adaptation method that we proposed, consisting of three domain adaptations of image level, feature level, and output level, is clear and well-structured; however, it is relatively complex. To increase the reproducibility, we have made the source code openly available at data.gpcv.whu.edu.cn.

Another issue is about the tradeoff hyperparameters in our method, i.e., the weights λ of different loss functions. There are several ways to determine these tradeoff parameters to make up a suitable combination. Several domain adaptation methods [5]–[10] based on adversarial learning have been proposed to deal with this. Some of them conducted experiments to observe the impact of the changes in tradeoff parameters, and others set tradeoff parameters empirically. We applied the parameters setting of SIFA [7] for image- and feature-level domain constraints and that of AdaptSegNet [8] for output-level domain constraints to our method, which are found feasible and stable in both data sets. However, in the future, we will try some automatic search strategies to further increase the automation level of the algorithm.

V. Conclusion

In this article, we proposed an end-to-end GAN-based full-space domain adaptation method for land cover classification of new remote sensing images with available GIS maps that are outdated or from other areas. In our method, the image, feature, and output space domain adaptations in two stages are fully utilized to align the source and target images and output the final segmentation maps simultaneously. Our experimental results demonstrated that our method exceeded the recent GAN-based UDA methods and showed the effectiveness of UDA in multisource remote sensing image classification. Our method also was shown to be a general framework that can be implemented for other domain transfer methods based on adversarial learning to boost their performance.

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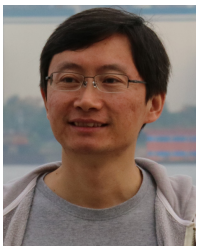
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